OUTPUT

Problem 1: Omitted Variable Bias

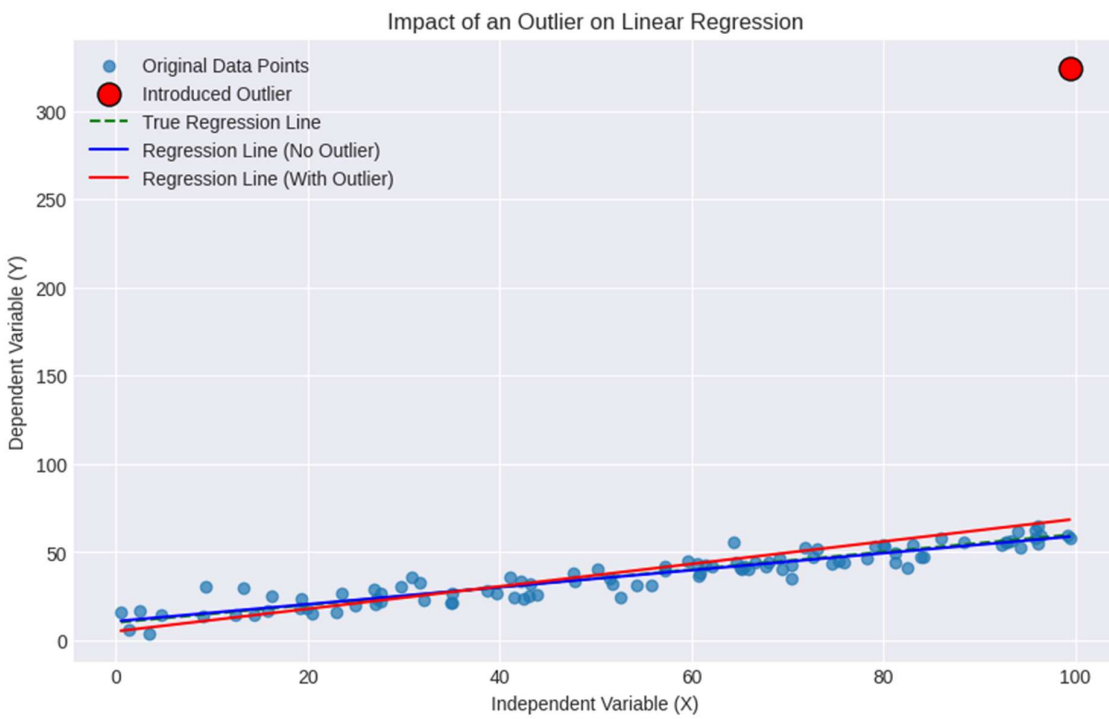
--- Running simulation with N=100 (Small Sample) ---
True coefficient for X (b_true): 2
Estimated coefficient for X (Full Model, N=100): 2.1192
Estimated coefficient for X (Omitted Model, N=100): 3.3990

--- Running simulation with N=10000 (Large Sample) ---
True coefficient for X (b_true): 2
Estimated coefficient for X (Full Model, N=10000): 2.0014
Estimated coefficient for X (Omitted Model, N=10000): 3.5061

Problem 2: Sensitivity to Outliers

--- Baseline Model (No Outliers) ---
True Intercept (Beta0): 10, True Slope (Beta1): 0.5
Estimated Intercept: 10.7298, Estimated Slope: 0.4824
R-squared: 0.8794

--- Model with Outlier ---
Estimated Intercept: 4.9951, Estimated Slope: 0.6370
R-squared: 0.3088



Problem 3: Model Selection

Loaded data (hardcoded) for Problem 3:

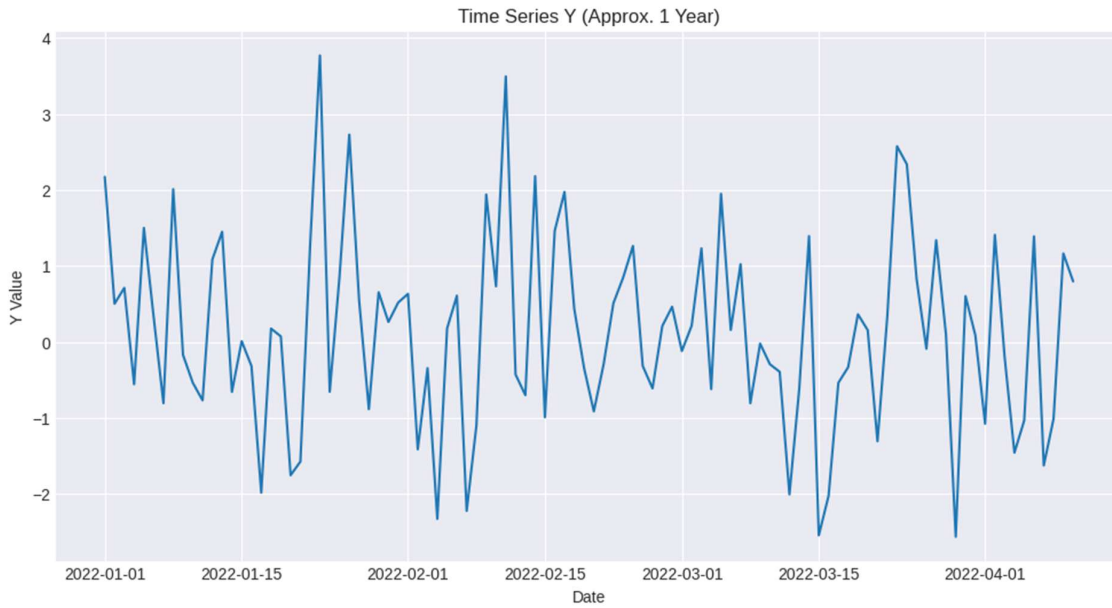
```
         Y      Z1        Z2       Z3       Z4       Z5
0  2.172296  0.121634 -0.051562  0.570616  1.279931  0.075233
1  0.502380  0.025446 -0.093062  0.304875 -0.582292  0.377388
2  0.711362 -0.136716 -0.082229 -0.191680 -0.647970  1.230986
3 -0.557168 -0.284459 -0.170922 -0.853670 -1.256146 -0.991686
4  1.500199  0.105205 -0.169141  0.826558  0.640945  1.099873

Dataset columns after renaming: ['y', 'x1', 'x2', 'x3', 'x4', 'x5']

--- Model Selection: Backward Elimination (Optimizing Adjusted R-squared) ---
Initial Full Model: Predictors ['x1', 'x2', 'x3', 'x4', 'x5'], Adjusted R-squared: 0.9936
  No further improvement in Adjusted R-squared by removing any predictor.

Final Best Model by Backward Elimination (Adjusted R-squared): ['x1', 'x2', 'x3', 'x4', 'x5']

--- Model Selection: Forward Selection (Minimizing AIC) ---
Initial Null Model: Predictors [], AIC: 334.0103
  Added 'x4'. New AIC: 141.1442. Current Predictors: ['x4']
  Added 'x3'. New AIC: 24.4665. Current Predictors: ['x4', 'x3']
  Added 'x5'. New AIC: -69.4350. Current Predictors: ['x4', 'x3', 'x5']
  Added 'x2'. New AIC: -118.1856. Current Predictors: ['x4', 'x3', 'x5', 'x2']
  Added 'x1'. New AIC: -165.9022. Current Predictors: ['x4', 'x3', 'x5', 'x2', 'x1']

Final Best Model by Forward Selection (AIC): ['x4', 'x3', 'x5', 'x2', 'x1']
```

Problem 4: Elasticity Calculations

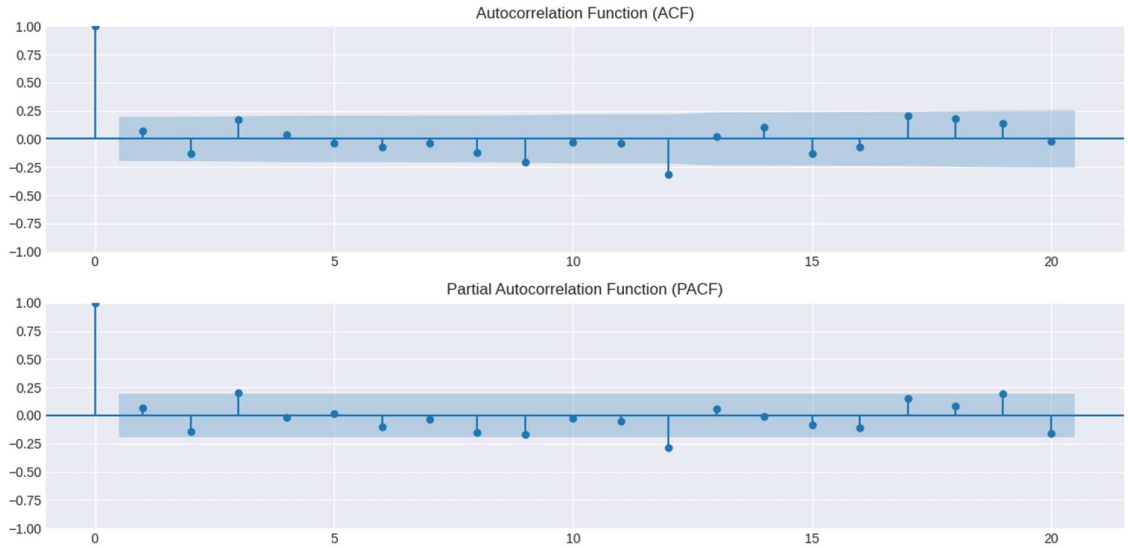
```
--- Problem 4a) Elasticity Formulas and Calculations ---

--- Example Calculations for Problem 4a (using x=5 as an example where applicable) ---
For x = 5:
  (a) y = 2 + 0.8x: Elasticity = 0.6667 (depends on x)
  (b) ln(y) = 0.1 + 0.4x: Elasticity = 2.0000 (depends on x)
  (c) ln(y) = 0.1 + 0.25ln(x): Elasticity = 0.2500 (constant)
  (d) y = 0.15 + 1.2ln(x): Elasticity = 0.5766 (depends on y, which depends on x)

--- Problem 4b) Identifying the Correct Regression for Constant Elasticity ---
The question asks: 'What is the elasticity of y with respect to x in this case? Hint.. You must
identify the correct regression to use for the purpose among the 4 given in the problem.'

**Analysis:**
We are looking for the model where the elasticity of y with respect to x is constant, implying it
doesn't change with the values of x or y.
Let's re-examine the elasticity formulas derived for each model:
  - (a) E_a = 0.8 * (x / (2 + 0.8x)): NOT constant, depends on x.
  - (b) E_b = 0.4x: NOT constant, depends on x.
  - (c) E_c = 0.25: CONSTANT. The elasticity is simply the coefficient of ln(x).
  - (d) E_d = 1.2/y: NOT constant, depends on y.
```

Conclusion for Problem 4b:
The only regression model among the four that yields a constant elasticity of y with respect to x is **model (c): ln(y) = 0.1 + 0.25ln(x)**.
In this specific case, the elasticity of y with respect to x is **0.25**.
This type of model (log-log) is often used in economics precisely because the coefficient of the logged independent variable directly represents a constant elasticity.

Problem 5: Time Series Analysis and Forecasting



--- ADF Test for Stationarity ---
ADF Statistic: -4.961
p-value: 0.000
Critical Values:
  1%: -3.500
  5%: -2.892
  10%: -2.583
Result: The series is stationary (reject H0)
Series is already stationary (or became stationary with d=0).

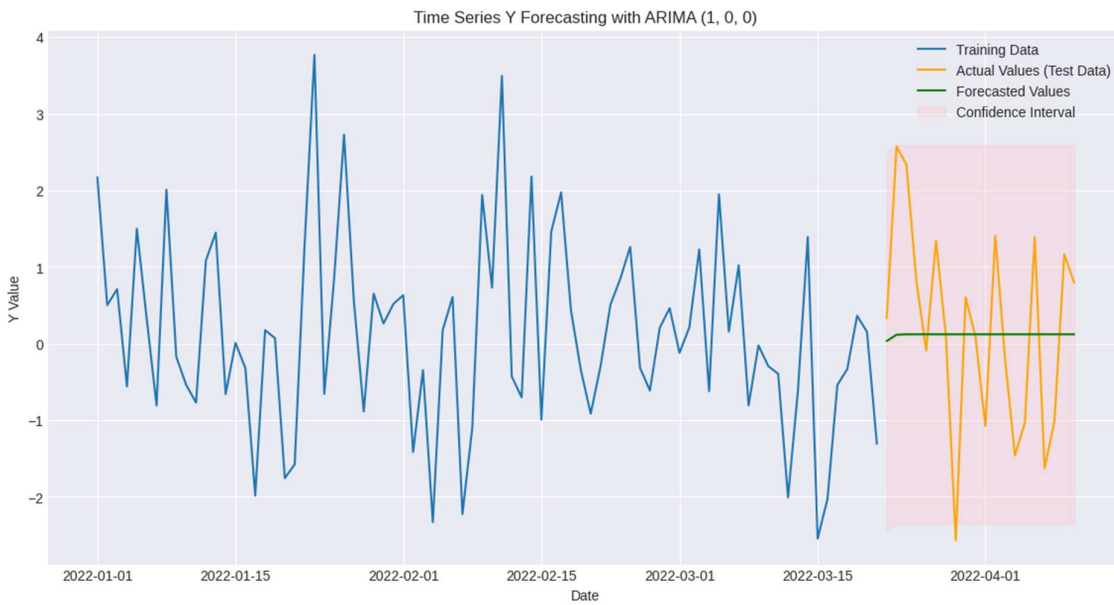


--- Manually Chosen ARIMA Order (p,d,q): (1, 0, 0) ---
Note: p and q are chosen manually. Inspect the ACF/PACF plots above to refine these values if necessary.
Given the problem description and lack of auto_arima, (1, d, 0) is a reasonable default for many financial series.

--- Fitting ARIMA Model with order (1, 0, 0) ---

```
                               SARIMAX Results
==============================================================================
Dep. Variable:              y_values   No. Observations:                   80
Model:                 ARIMA(1, 0, 0)   Log Likelihood                -131.872
Date:                Mon, 09 Jun 2025   AIC                            269.744
Time:                        13:36:33   BIC                            276.891
Sample:                    01-01-2022   HQIC                           272.609
                         - 03-21-2022
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.1222      0.154      0.793      0.428      -0.180       0.424
ar.L1          0.0616      0.131      0.471      0.638      -0.195       0.318
sigma2         1.5823      0.248      6.375      0.000       1.096       2.069
===================================================================================
=====
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):                 2.02
Prob(Q):                              0.97   Prob(JB):                         0.36
Heteroskedasticity (H):               0.62   Skew:                             0.36
Prob(H) (two-sided):                  0.22   Kurtosis:                         3.27
===================================================================================
=====

Warnings:
```

[1] Covariance matrix calculated using the outer product of gradients (complex-step).
/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/base/tsa_model.py:473:
ValueWarning: No frequency information was provided, so inferred frequency D will be used.
  self._init_dates(dates, freq)
/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/base/tsa_model.py:473:
ValueWarning: No frequency information was provided, so inferred frequency D will be used.
  self._init_dates(dates, freq)
/usr/local/lib/python3.11/dist-packages/statsmodels/tsa/base/tsa_model.py:473:
ValueWarning: No frequency information was provided, so inferred frequency D will be used.
  self._init_dates(dates, freq)



RMSE (Root Mean Squared Error) on Test Data: 1.323

Problem 6: Structural Break

--- Hardcoded Data for Problem 6 (20 periods with simulated structural break) ---

	t	Y	X	D	X_D
0	1	11.962363	10.993428	0	0.000000
1	2	9.635959	9.723471	0	0.000000
2	3	10.715217	11.295377	0	0.000000
3	4	10.098282	13.046060	0	0.000000
4	5	9.221464	9.531693	0	0.000000
5	6	9.876786	9.531726	0	0.000000
6	7	10.428219	13.158426	0	0.000000
7	8	11.143133	11.534869	0	0.000000
8	9	8.929887	9.061051	0	0.000000
9	10	10.250866	11.085120	0	0.000000
10	11	18.008040	9.073165	1	9.073165
11	12	20.455089	9.068540	1	9.068540
12	13	20.712390	10.483925	1	10.483925
13	14	13.202448	6.173440	1	6.173440
14	15	15.647791	6.550164	1	6.550164

```
15  16  17.092294   8.875425  1   8.875425
16  17  17.170370   7.974338  1   7.974338
17  18  18.983072  10.628495  1  10.628495
18  19  15.947742   8.183952  1   8.183952
19  20  15.959950   7.175393  1   7.175393

======================================================================

--- Regression Analysis for Problem 6a) ---
Model Formulated: Y(t) = α + β1*X(t) + γ*D(t) + δ*(X(t)*D(t)) + η(t)
Equivalent Statsmodels Formula: Y ~ X + D + X_D
                            OLS Regression Results
==============================================================================
Dep. Variable:                      Y   R-squared:                       0.946
Model:                            OLS   Adj. R-squared:                  0.936
Method:                 Least Squares   F-statistic:                     93.04
Date:                Mon, 09 Jun 2025   Prob (F-statistic):           2.43e-10
Time:                        13:36:33   Log-Likelihood:                -26.539
No. Observations:                  20   AIC:                             61.08
Df Residuals:                      16   BIC:                             65.06
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      6.7203      2.582      2.603      0.019       1.247      12.193
X              0.3218      0.235      1.369      0.190      -0.177       0.820
D             -0.6802      3.218     -0.211      0.835      -7.503       6.142
X_D            1.0179      0.325      3.128      0.006       0.328       1.708
==============================================================================
Omnibus:                        3.082   Durbin-Watson:                   2.113
Prob(Omnibus):                  0.214   Jarque-Bera (JB):                1.890
Skew:                           0.752   Prob(JB):                        0.389
Kurtosis:                       3.087   Cond. No.                         186.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

--- Explanation of Testing for Structural Break (Problem 6a) ---
The approach involves using a single regression model with an added dummy variable and an
interaction term to capture potential shifts in both the intercept and the slope.

**1. Design of the Regression Model:**
The model estimated is: Y ~ X + D + X_D
  - `Y`: The dependent variable.
  - `X`: The independent variable.
  - `D`: A dummy variable (0 for periods before/at the breakpoint, 1 for periods after).
```

- `X_D`: An interaction term, calculated as `X * D`.

2. Interpretation of Coefficients:
- The coefficient of `X` (named `X` in the summary) represents `β1`, the slope for the initial period (t=1 to t=10).
- The coefficient of `D` (named `D` in the summary) represents `γ`, the change in the intercept after the breakpoint.
- The coefficient of `X_D` (named `X_D` in the summary) represents `δ`, the *change* in the slope (`β`) after the breakpoint.
- The `Intercept` coefficient represents `α`, the intercept for the initial period.

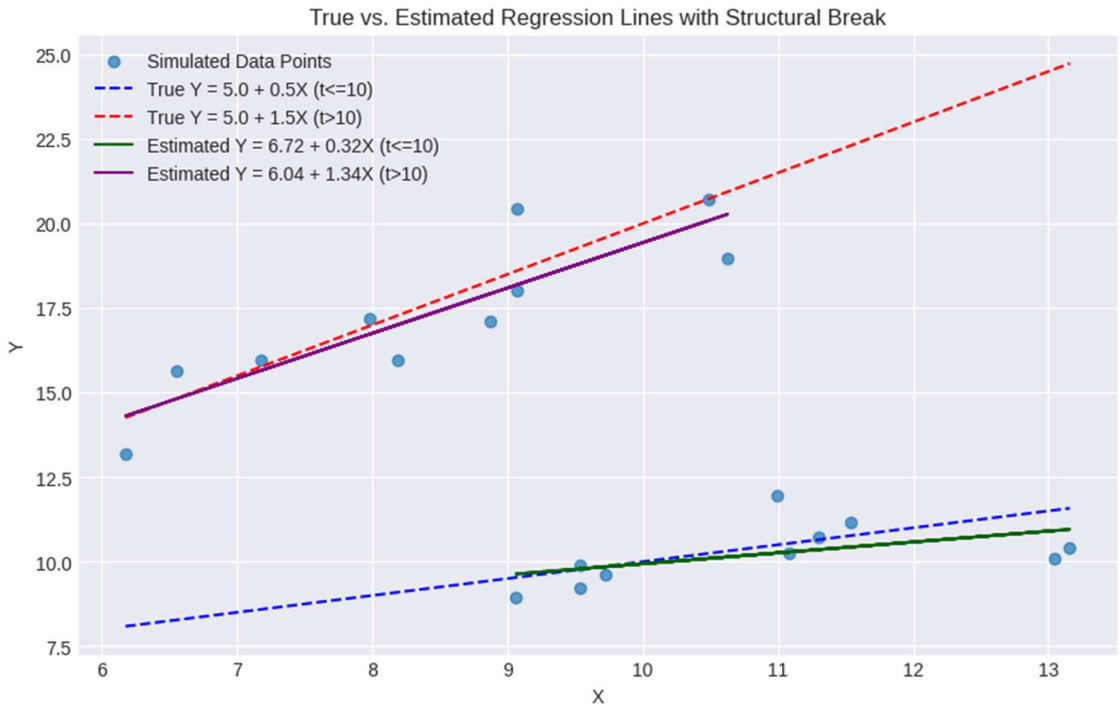
3. Testing for Structural Break in β:
To test if the slope (`β`) changes at the breakpoint (t=10), we look at the statistical significance of the `X_D` coefficient (δ).
- **Null Hypothesis (H0):** The change in slope (δ) is zero, meaning `β1 = β2` (no structural break in the slope).
- **Alternative Hypothesis (H1):** The change in slope (δ) is not zero, meaning `β1 ≠ β2` (a structural break in the slope exists).

In our simulated data, the true initial slope (β1) was 0.5 and the true post-break slope (β2) was 1.5.
Therefore, the true change in slope (δ) was 1.00.
From the regression summary, the estimated coefficient for 'X_D' (δ_hat) is 1.0179 with a p-value of 0.0065.

Conclusion: Since the p-value for 'X_D' is less than 0.05, we **reject the null hypothesis (H0)**.
This indicates statistically significant evidence that the β parameter (slope) *did* undergo a structural break at t=10 in this simulated dataset.

Estimated Parameters based on the single regression model:
- Intercept (first regime): 6.7203 (True: 5.0)
- Slope (first regime, β1): 0.3218 (True: 0.5)
- Change in Intercept (γ): -0.6802
- Change in Slope (δ): 1.0179 (True: 1.00)

Inferred parameters for the second regime (t > 10):
- Inferred Intercept (second regime): 6.0401
- Inferred Slope (second regime, β2): 1.3396 (True: 1.5)


True vs. Estimated Regression Lines with Structural Break
Simulated Data Points
True Y = 5.0 + 0.5X (t<=10)
True Y = 5.0 + 1.5X (t>10)
Estimated Y = 6.72 + 0.32X (t<=10)
Estimated Y = 6.04 + 1.34X (t>10)
25.0
22.5
20.0
17.5
15.0
12.5
10.0
7.5
Y
6
7
8
9
10
11
12
13
X